

QuantumTrade19 — Module 01 Gap-Closure — FINAL LOCKED SUMMARY

Module: 01 — UI/UX Design System & App Shell (Gap-Closure Round)

Base version: v0.1.9-module01-locked

This round closes on top of that base and should be tagged/versioned as the next increment (e.g. v0.1.10-module01-gapclosure-locked) once you commit.

Status: ALL ITEMS LOCKED — this round is fully closed.

Date locked: August 15, 2026

1. Purpose of This Round

Module 01 was previously locked at v0.1.9, but a finalized gap-closure plan identified 11 specific gaps left open from the original build (security hardening, UI polish, missing wiring between built-but-disconnected Workers, and one net-new feature). This document records exactly what was built, what was deliberately left alone as a confirmed decision, what was explicitly deferred to later modules, and every bug discovered and fixed along the way — so this is the single reference point when the full software's final blueprint gets assembled.

2. Full Gap-Closure Progress Table

#	Item	Status	Notes
1	Argon2 password hashing (replaces SHA-256)	✔ Locked	Breaking change — old accounts had to re-register
—	<i>(discovered)</i> Register button missing on Login screen	✔ Locked	Bugfix, not originally planned
2	SecurityMonitor.get_setting/set_setting generic passthrough	✔ Locked	Named methods left fully intact
3	get_per_symbol_setting/set_per_symbol_setting interface stubs	✔ Locked	Interface-only; real exercise deferred to Module 02
4	data/settings.json stays plain JSON (not encrypted)	✔ Confirmed decision	No build needed — settings-only, not secrets
5	CSS cross-fade for theme switching	✔ Locked	
—	<i>(discovered)</i> Pre-login screens missing --qt19-* theme CSS vars	✔ Locked	Bugfix — vars were only set inside the authenticated shell

#	Item	Status	Notes
6	Sound Engine wired to real placeholder tones	✔ Locked	6 programmatically-generated .wav tones; swappable later
—	(<i>discovered</i>) Square hover-shadow on radius-full buttons	✔ Locked	Bugfix — Radix internal state-layer radius mismatch
7	Animation_Choreographer_Worker / Page_Transition_Worker → real CSS	✔ Locked	Selection logic moved out of AppState into the Workers
8	Tab-to-tab switch animation	✔ Locked	Extended mid-build to be independently Settings-controlled
9	Custom cursor stays removed (plain OS cursor)	✔ Confirmed decision	No build needed
10	"Remember this device" 60-day TOTP-skip toggle	✔ Locked	Trust stored via OS keyring, machine-bound
11	Shake animation + error sound on failed login	✔ Locked	Built immediately after item 6, per plan
—	Tray icon / minimize-to-tray	Deferred → Module 19	Do not attempt before then
—	Full per-symbol-settings testing	Deferred → Module 02	Interface exists (item 3); no real symbols to test against yet

3. What Was Built, Item by Item

Item 1 — Argon2 Password Hashing

- **Files:** engines/workers/security/auth_login_worker.py
- Replaced raw SHA-256 + manual salt with argon2-cffi's PasswordHasher. Argon2 embeds its own random salt and parameters inside the returned hash string — the old manual salt-concatenation approach was removed entirely.
- Added check_needs_rehash auto-upgrade path so future parameter changes re-hash transparently on next successful login.
- **Breaking change, as flagged at the time:** any account hashed under the old SHA-256 format could no longer authenticate. Required re-registration.
- **Follow-on bug found:** the Login screen had no path to reach Registration except on first run — added go_to_register() to MasterAppEngine and AppState, plus a "New here? Create an account" link on Login. This became the practical fix that let re-registration actually happen.

Item 2 — Generic Settings Passthrough

- **Files:** `engines/monitors/security_monitor.py`
- Added `get_setting(key, default) / set_setting(key, value)`, calling straight through to `SettingsPersistenceWorker.load()/save()`. Every existing named accessor (`is_totp_enabled`, `is_telegram_enabled`, etc.) untouched.

Item 3 — Per-Symbol Setting Interface

- **Files:** `engines/workers/security/settings_persistence_worker.py`, `engines/monitors/security_monitor.py`
- Added `per_symbol_settings: {}` to persistence defaults, plus `get_per_symbol_setting/set_per_symbol_setting` on both the Worker and the Monitor passthrough. Symbol is currently a free-text key since no `Symbol_Registry_Worker` exists yet — Module 02 will exercise this for real.

Item 4 — settings.json Stays Plain JSON

- Confirmed decision, no code change. Settings-only data, not secrets — secrets already route through the OS keyring via `SecureKeyStorageWorker`.

Item 5 — CSS Cross-Fade for Theme Switching

- **Files:** `ui/theme/glass.py`, `quantumtrade19/quantumtrade19.py`
- Added a uniform `_THEME_CROSSFADE` transition (background, border-color, box-shadow, color, 0.9s ease) to every style dict driven by `--qt19-* vars: GLASS_CARD_STYLE, GLASS_CARD_3XL_STYLE, PILL_BUTTON_STYLE, GLOW_RING_STYLE, PAGE_BG_STYLE`.
- **Bug found mid-build:** the `--qt19-*` custom properties were only ever set inside `page_shell.py` (the authenticated shell), so every pre-login screen (splash, login, register, forgot-password, manage-security, app-lock) never actually received themed colors — they were rendering on invalid/fallback custom-property values the whole time. Fixed by moving the variable binding to the app-root wrapper in `quantumtrade19.py`, which wraps every screen.

Item 6 — Sound Engine Wired to Real Audio

- **Files:** `engines/workers/ui_experience/sound_engine_worker.py`, `engines/monitors/ui_experience_monitor.py`, `state/app_state.py`, `standalone/generate_placeholder_sounds.py`
- Generated 6 placeholder sine-wave tones (click, page_change, tab_slide, card_flip, error, success) into `assets/sounds/*.wav` via a one-time local script.
- `SoundEngineWorker.play(event_name)` resolves an event name to a clip URL, gated by the master mute setting. `UIExperienceMonitor.play_sound()` exposes it upward. `AppState.play_sound()` triggers actual browser playback via `rx.call_script + new Audio().play()`, since Python has no direct browser audio access.

- **Bug found mid-build:** a squared hover-glow appeared behind pill-shaped buttons (Logout/Cancel) in light themes. Root cause: Radix Themes' internal hover/press state-layer overlay used its own corner-radius token instead of inheriting the button's `radius="full"` prop. Fixed with a global `[data-radius="full"] { overflow: hidden !important; }` rule in `global_css.py`, clipping any mismatched internal overlay to the button's own rounded shape — a general fix protecting every full-radius button app-wide, not just the one dialog.

Item 7 — Connect Animation_Choreographer_Worker / Page_Transition_Worker

- **Files:** `engines/workers/ui_experience/animation_choreographer_worker.py`, `engines/workers/ui_experience/page_transition_worker.py`, `state/app_state.py`
- These two Workers previously existed only as empty stubs while the real single/sequential/shuffle transition-selection logic lived inline inside AppState, completely bypassing them. Moved that logic into `PageTransitionWorker.pick()`, with `AnimationChoreographerWorker` now owning the canonical list of transition effect names and their CSS class mapping. `AppState._pick_transition_effect()` now routes through the engine instead of reimplementing the pool logic.
- Also added `AnimationChoreographerWorker.error_preset()` returning the `qt19-shake` class name, formally connecting the "error preset" language from the original spec to the actual shake animation (item 11).

Item 8 — Tab-to-Tab Switch Animation

- **Files:** `ui/theme/global_css.py`, `quantumtrade19/quantumtrade19.py`, `state/app_state.py`, `engines/workers/security/settings_persistence_worker.py`, `ui/pages/settings.py`
- Initially built as a fast, fixed fade+slide (`qt19-tab-switch`, 0.35s) applied via the same remount-via-key technique used for screen transitions.
- **Extended per your request mid-build**, before locking: tab animation is now independently Settings-controlled, reusing the same 10-effect catalog and single/sequential/shuffle mode pattern as the screen-entrance transitions, but with its own separate pool/mode/persisted state (`tab_transition_effects_enabled`, `tab_transition_mode`) — fully independent from the Login→Dashboard transition settings. New "Tab Switch Animation" card added to the Settings page.
- Also wired the previously-unused `tab-slide` sound (from item 6) into every tab click.

Item 9 — Custom Cursor Stays Removed

- Confirmed decision, no code change. `qt19_cursor_glow()` is a deliberate no-op (verified during bug investigation — see below), restoring the plain OS cursor everywhere it's still imported for backward compatibility.

Item 10 — "Remember This Device" 60-Day Toggle

- **Files:** `engines/monitors/security_monitor.py`, `engines/masters/master_app_engine.py`, `ui/pages/login.py`, `state/app_state.py`
- Added `is_device_trusted()` / `trust_this_device()` / `clear_device_trust()` to `SecurityMonitor`, storing a `device_trust_until` epoch timestamp via the OS keyring (`SecureKeyStorageWorker`) — machine-bound by the same mechanism as the TOTP secret, so trust never crosses devices.
- `SecurityMonitor.login()` now skips the TOTP check entirely when the device is currently trusted, regardless of whether TOTP is enabled.
- Username + password remain always required — only TOTP is skippable, per your explicit confirmation.
- **Scope note recorded at build time:** `login()` is shared by both `attempt_login` (initial login) and `unlock()` (App Lock re-entry), so trusting a device also skips TOTP on App Lock unlock. This was a deliberate architectural choice (splitting the code paths for no clear benefit was judged worse) — flag it if App Lock should behave differently later.
- Added the "Remember this device (skip 2FA for 60 days)" checkbox on Login, shown only when the TOTP field itself is visible.

Item 11 — Shake Animation + Error Sound on Failed Login

- **Files:** `ui/theme/global_css.py`, `engines/workers/ui_experience/animation_choreographer_worker.py`, `ui/pages/login.py`, `state/app_state.py`
- Added the `qt19-shake` keyframe (0.4s horizontal shake). `submit_login()` now increments `login_error_seq` on both failure branches (empty fields, wrong credentials), forcing a fresh remount via `key=` so the shake replays on every consecutive failed attempt, not just the first. Error sound (from item 6) plays on the same failure branches.

4. Bugs Found and Fixed During This Round

These were not part of the original 11-item plan but were discovered and resolved while building it. Recorded here for the historical record and to prevent regression in later modules.

1. **Missing Register path from Login.** No way to reach the Registration screen except on first-run — critical since item 1's Argon2 migration required re-registration. Fixed with `go_to_register()` end-to-end.
2. **Pre-login screens missing theme CSS variables.** `--qt19-*` custom properties were scoped only to the authenticated shell, silently breaking themed glass styling on `Splash/Login/Register/Forgot-Password/Manage-Security/App-Lock`. Fixed by moving the binding to the app root.
3. **Square hover-shadow on rounded buttons.** Radix Themes' internal state-layer overlay ignored the button's own border-radius. Fixed with a global overflow-clipping CSS rule tied to Radix's `data-radius="full"` attribute.

4. **Recurring stale-frontend-bundle issue.** Reflex's dev-server hot-reload does not reliably pick up brand-new State methods — repeatedly caused "nothing happens" symptoms (sound not playing, buttons appearing to do nothing) that were actually just old cached bundles. Resolution pattern established: full server restart (`Ctrl+C + reflex run`) + hard browser refresh (`Ctrl+Shift+R`) after any new state method or root-file change — this should be treated as standard procedure for all future modules, not just this one.
5. **Whitespace corruption when large files pass through chat.** Several multi-hundred-line files lost indentation when relayed through this conversation, causing at least one real `IndentationError (settings_persistence_worker.py)`. Established practice going forward: for files larger than ~100 lines, prefer the user pasting the actual current file content directly (preserving real indentation) over reconstructing from a lossy transcript.

5. File Manifest — Every File Touched This Round

Security / Engines:

- `engines/workers/security/auth_login_worker.py`
- `engines/workers/security/settings_persistence_worker.py`
- `engines/monitors/security_monitor.py`
- `engines/masters/master_app_engine.py`

UI Experience Engines:

- `engines/workers/ui_experience/sound_engine_worker.py`
- `engines/workers/ui_experience/animation_choreographer_worker.py`
- `engines/workers/ui_experience/page_transition_worker.py`
- `engines/monitors/ui_experience_monitor.py`

Theme / Global CSS:

- `ui/theme/glass.py`
- `ui/theme/global_css.py`

Pages / Components:

- `ui/pages/login.py`
- `ui/pages/settings.py`
- `ui/components/logout_dialog.py`

State / App Entry:

- `state/app_state.py`
- `quantumtrade19/quantumtrade19.py`

Standalone / One-Time Scripts:

- `generate_placeholder_sounds.py` (run once locally; not part of the running app)

New Assets:

- `assets/sounds/click.wav`, `page_change.wav`, `tab_slide.wav`, `card_flip.wav`, `error.wav`, `success.wav`

6. Forward Notes for the Final Blueprint

- **Module 02 owes two things back to this module:** (a) real exercise of the per-symbol settings interface (item 3) once `Symbol_Registry_Worker` exists, and (b) it should be the first place that actually calls `SecurityMonitor.get_per_symbol_setting/set_per_symbol_setting` for something real, validating the interface built here.
- **Module 19 owes:** tray icon / minimize-to-tray, deferred cleanly with zero UI Shell dependencies blocking it.
- **App Lock + "Remember this device" interaction** (see item 10 scope note) should be revisited once App Lock's UX is finalized — decide explicitly whether a trusted device should also skip TOTP on unlock, or whether that needs its own separate trust flag.
- **The sound placeholder tones are intentionally simple** and explicitly designed to be swapped for branded audio later — swapping is a pure asset replacement (same filenames in `assets/sounds/`), no code changes required.
- **Established engineering conventions from this round**, worth carrying forward into every future module: full-server-restart + hard-refresh after any state/root-file change; prefer direct file pastes over transcript reconstruction for large files; the remount-via-key technique (screen transitions, tab switches, login shake) is now the standard pattern for any future one-shot CSS animation tied to a Reflex state change.

Module 01 (UI/UX & App Shell) — Gap-Closure Round: CLOSED.